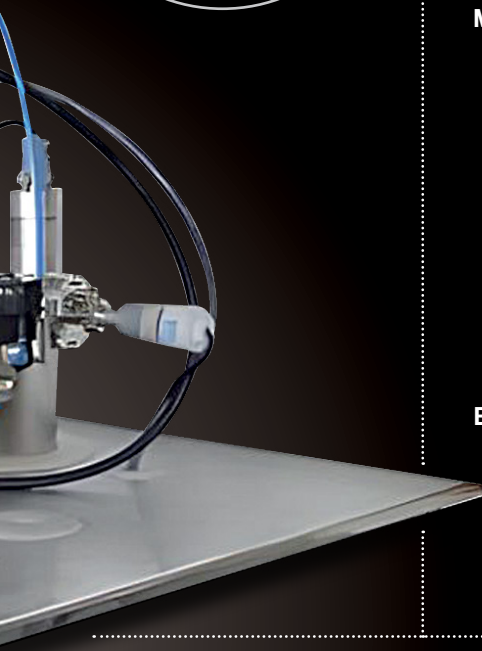
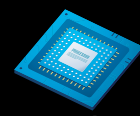
₹8 LAKH

Price of a 6 ct lab diamond. An equivalent natural diamond could cost 10 times as much



PRECIOUS STONE

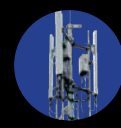
Given their unique chemical, thermal and mechanical properties, high-purity lab diamonds find use in...



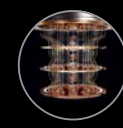
Microchips



Satellites



5G/ 6G networks



Quantum computing



Biomedical implants

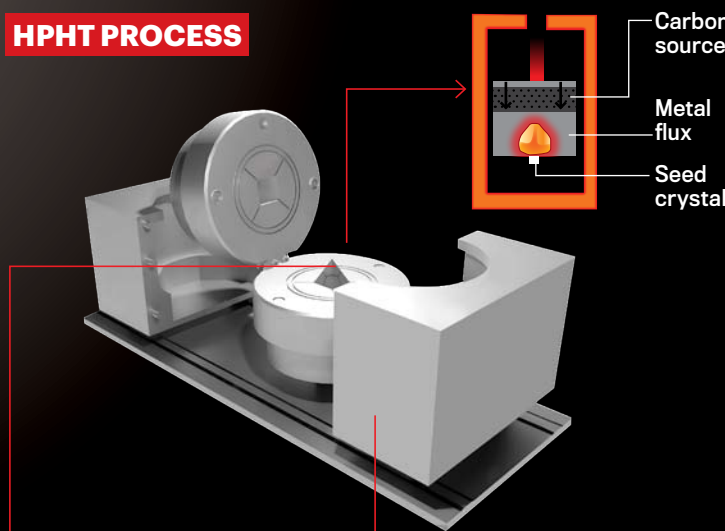


Defence

Low-quality LGDs are used for protective coatings on cutting tools

Two key technologies are used globally to grow LGDs: Chemical Vapour Deposition (CVD) & High Pressure High Temperature (HPHT)

HPHT PROCESS



- Diamond seed is placed in a growth chamber along with a carbon source (graphite) and catalyst metal (iron, nickel or cobalt)
- The chamber is heated to 1,300-1,600°C with intense pressure of 1.5 million pounds per sq. inch
- The molten metal flux dissolves the carbon source
- Carbon atoms migrate through the flux towards the cooler diamond seed and crystallise on it
- A synthetic diamond begins to grow



"LGD IS A TECHNOLOGY AND INNOVATION-DRIVEN EMERGING SECTOR THAT HAS A HIGH EMPLOYMENT POTENTIAL"

NIRMALA SITHARAMAN
Union Finance Minister
in her 2023 Budget speech

in sparkling colours besides the pristine colourless and transparent variety, they are up to 90 per cent cheaper than mined diamonds of similar size and more ethical in their manufacturing process. The world's biggest jewellers and luxury watchmakers are bedazzled, as are diamantaires in India.

HOW TO GROW A DIAMOND

The diamond cultivation technology is nothing new; it has excited researchers and businesses for over a century now. The first LGDs, developed in the 1950s, were produced for industrial use by General Electric. Gem-quality diamonds followed 20 or so years later, but their production on a commercial scale is a phenomenon that took off in the first decade of this century.

Two key technologies are in use currently to grow diamonds in the laboratory. Chinese firms mainly use the costlier High Pressure High Temperature or HPHT method, in which a wafer-thin diamond segment, called the seed crystal, is placed in a growth chamber along with a metal catalyst and a carbon source (see A Cut Above). Intense pressure of 1.5 million PSI and a high temperature of 1,300-1,600°C melt the metal and carry the carbon atoms from the source to the seed. These atoms begin to accumulate on the seed, layer by layer, slowly building up into a larger diamond replicating its crystal structure.

Firms in India, on the other hand, use